39_Congo

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1 How have female and male primary school repetition rates evolved relative to one another in Congo between 1972 and 2012?

1.1 Abstract

Using World Bank World Development Indicators (WDI), this study examines the evolution of male and female primary school repetition rates in Congo between 1972 and 2012. The analysis compares two key indicators: the percentage of female students repeating primary grades and the percentage of male students repeating primary grades. Over this forty-year period, repetition rates for both genders followed very similar trajectories, with male rates consistently slightly higher than female rates throughout. Despite this small gender gap, the overall pattern was marked by strong volatility — with pronounced fluctuations and sudden shifts in repetition levels over time. The data reveal a significant U-shaped decline between 1975 and 1985, followed by a period of sharp increases and irregular oscillations until around 2000, when repetition rates dropped abruptly. From 2000 onward, both male and female repetition rates displayed moderate volatility but an overall downward trend. These results suggest persistent structural challenges within the primary education system, despite gradual improvements in reducing grade repetition toward the end of the period.

1.2 1. Question

How have female and male primary school repetition rates evolved relative to one another in Congo between 1972 and 2012?

- **Female repetition proxy**: Repeaters, primary, female (% of female enrollment)
- **Male repetition proxy**: Repeaters, primary, male (% of male enrollment)

1.3 2. Data

- **Source**: World Bank World Development Indicators (WDI)
- **Indicators**:
  - Repeaters, primary, female (% of female enrollment)
  - Repeaters, primary, male (% of male enrollment)
- **Coverage**: Congo, 1972–2012
- **Notes**: National-level data only

1.4 3. Method

1. Filtered dataset for Congo and selected the two primary repetition rate indicators.
2. **Extracted relevant columns**: Year, Indicator Name, and Value.
3. Pivoted the dataset to create a side-by-side comparison of male and female repetition rates between 1972 and 2012.
4. Generated a dual-line time series plot to visualize gender differences and overall volatility in repetition trends.

(Analysis is descriptive; no causal inference applied.)

1.5 4. Results

- **Female repetition rates**: Showed a volatile trajectory between 1972 and 2012, with significant ups and downs, but an overall moderate downward trend toward the end of the period.
- **Male repetition rates**: Mirrored female repetition rates closely, also exhibiting strong volatility and sudden shifts, while remaining consistently slightly higher than female rates.
- **Comparison**: The two indicators moved largely in parallel over time, with males slightly above females throughout. Both experienced a pronounced U-shaped drop between 1975 and 1985, followed by irregular increases and fluctuations until a sharp drop around 1999–2000, then moderate volatility with an overall downward trend to 2012.

(Figure 1. Congo: Female vs. Male Primary School Repetition Rates, 1972–2012)

(Table 1. Pivoted dataset summary)

1.6 5. Interpretation

- The near-identical trajectories for male and female repetition rates indicate that gender was not a major differentiating factor in primary school retention.
- The persistent volatility points to structural instability in Congo's primary education system — likely linked to political upheavals, funding fluctuations, and inconsistent policy implementation.
- The sharp drop around 2000 may reflect reforms in educational policy or data reclassification coinciding with post-conflict reconstruction efforts.
- The overall downward trend in the 2000s suggests gradual progress toward reducing grade repetition and improving educational efficiency.
- These findings highlight both the resilience of the education system and the need for more sustained interventions to maintain consistent learning outcomes over time.

1.7 6. Limitations

- The data are aggregated at the national level and do not capture regional or rural-urban differences in repetition rates.
- WDI estimates for earlier decades may rely on incomplete administrative records, introducing potential uncertainty in year-to-year variation.
- The descriptive approach does not isolate causal factors such as teacher quality, resource allocation, or curriculum reforms that may explain observed fluctuations.

1.8 7. Next Steps / Extensions

- Extend the analysis to include total repetition rates and enrollment levels to assess efficiency in the broader primary education system.
- Explore correlations between repetition rates and macro variables such as education expenditure, conflict intensity, or teacher–student ratios.
- Compare Congo's trajectory with neighboring Central African countries to identify regional convergence or divergence in educational performance.
- Conduct time-series decomposition to separate long-term trends from short-term volatility and structural breaks, particularly around major policy shifts.

[1]:
```python
# How have female and male primary school repetition rates evolved relative to␣
↪one another in Congo between 1972 and 2012?

import pandas as pd
import matplotlib.pyplot as plt
import os

# Folders
data_raw_folder = "data_raw/"
data_clean_folder = "data_clean/"
figures_folder = "figures/"

# Load CSV
filename = "gender_cog_filtered.csv"  # Filtered dataset with only relevant rows
df = pd.read_csv(os.path.join(data_raw_folder, filename))

# Keep only needed columns
df = df[["Year", "Indicator Name", "Value"]]

# Convert Year and Value to numeric, drop invalid rows
df["Year"] = pd.to_numeric(df["Year"], errors="coerce")
df["Value"] = pd.to_numeric(df["Value"], errors="coerce")
df = df.dropna(subset=["Year", "Value"])

# Pivot indicators into separate columns
df_pivot = df.pivot(index="Year", columns="Indicator Name", values="Value").
↪reset_index()
df_pivot = df_pivot.sort_values("Year")

print("Pivoted Congo dataset:")
display(df_pivot)

# Interpolate missing values for smooth plotting (optional)
df_plot = df_pivot.interpolate(method='linear')

# Plot the indicators
plt.figure(figsize=(10,6))
```

```
plt.plot(df_plot["Year"], df_plot["Repeaters, primary, female (% of female␣
↪enrollment)"],
         marker='o', linestyle='-', label="Repeaters, primary, female (% of␣
↪female enrollment)")
plt.plot(df_plot["Year"], df_plot["Repeaters, primary, male (% of male␣
↪enrollment)"],
         marker='o', linestyle='-', label="Repeaters, primary, male (% of male␣
↪enrollment)")

plt.title("Congo: Female vs Male primary repeaters (% of enrollment)␣
↪(1972-2012)")
plt.xlabel("Year")
plt.ylabel("Percentage")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.savefig(os.path.join(figures_folder,␣
↪"congo_female_vs_male_primary_repeaters.png"))
plt.show()

# Save cleaned CSV
df_pivot.to_csv(os.path.join(data_clean_folder,␣
↪"congo_female_vs_male_primary_repeaters"), index=False)
```

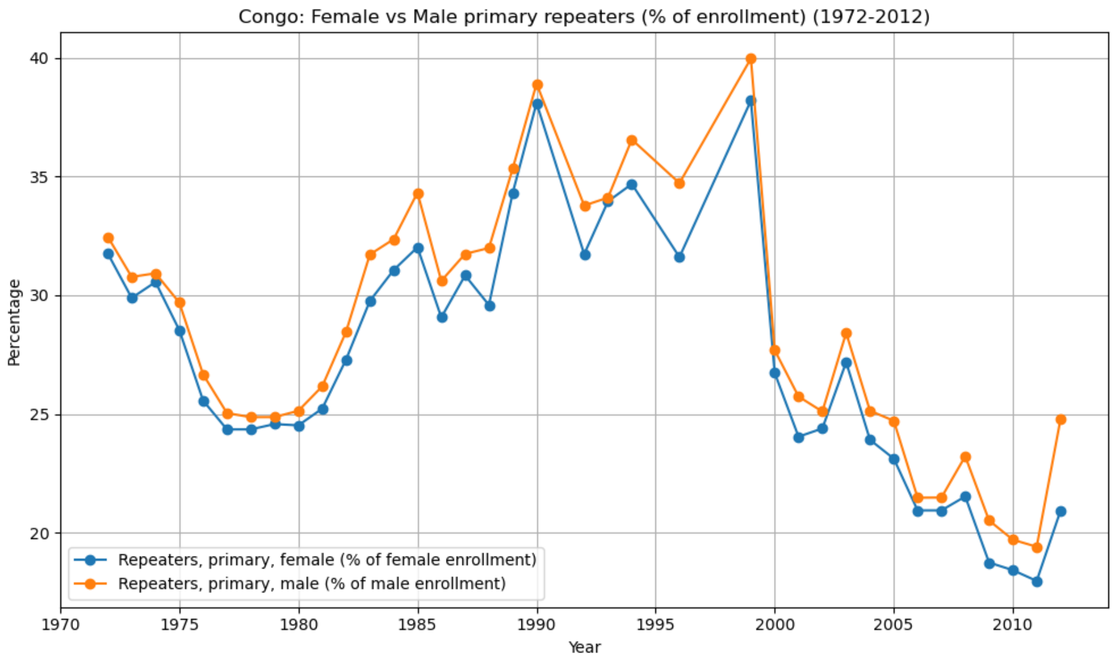
```
Pivoted Congo dataset:

Indicator Name  Year  Repeaters, primary, female (% of female enrollment)  \
0               1972                                           31.74897
1               1973                                           29.88408
2               1974                                           30.55378
3               1975                                           28.52698
4               1976                                           25.55829
5               1977                                           24.34761
6               1978                                           24.34955
7               1979                                           24.57212
8               1980                                           24.51837
9               1981                                           25.22356
10              1982                                           27.28710
11              1983                                           29.75602
12              1984                                           31.06046
13              1985                                           32.01797
14              1986                                           29.06406
15              1987                                           30.83229
16              1988                                           29.58489
17              1989                                           34.31414
18              1990                                           38.09336
19              1992                                           31.72920
20              1993                                           33.95043
```

21	1994	34.68253
22	1996	31.60380
23	1999	38.19295
24	2000	26.72443
25	2001	24.04016
26	2002	24.38775
27	2003	27.17869
28	2004	23.90940
29	2005	23.11565
30	2006	20.93691
31	2007	20.93692
32	2008	21.52317
33	2009	18.74102
34	2010	18.41952
35	2011	17.96675
36	2012	20.92145

Indicator Name	Repeaters, primary, male (% of male enrollment)
0	32.45162
1	30.77039
2	30.92216
3	29.68636
4	26.66065
5	25.03518
6	24.86360
7	24.86642
8	25.13242
9	26.16134
10	28.45654
11	31.71287
12	32.35739
13	34.31913
14	30.60149
15	31.73908
16	31.98704
17	35.33591
18	38.90709
19	33.77387
20	34.10978
21	36.55567
22	34.71913
23	39.96583
24	27.68104
25	25.73139
26	25.09964
27	28.41356
28	25.13263
29	24.71084

30	21.47057
31	21.47834
32	23.21660
33	20.51257
34	19.70110
35	19.40558
36	24.77563

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